Darshan UNIVERSITY

योग: कर्मसु कौशलम्

Python Programming - 2301CS404

Lab - 10

Roll No :495

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01) WAP to read and display the contents of a text file. (also try to open the file in some other directory)

- in the form of a string

- line by line

- in the form of a list

In [17]:
```
fp = open("demo.txt","r")

d1 = fp.read()
fp.seek(0)
d2 = fp.readline()
fp.seek(0)
d3 = fp.readlines()
fp.close()
print(d1)
print(d2)
print(d3)
```

```
Hello Students from the file
Dhruv kakkad
Heet
Hello Students from the file

['Hello Students from the file \n', 'Dhruv kakkad\n', 'Heet']
```

02) WAP to create file named "new.txt" only if it doesn't exist.

In [18]:
```
fp = open("new.txt","x")
data = "Hello World"
fp.write(data)
fp.close()
```

03) WAP to read first 5 lines from the text file.

In [24]:
```
fp = open("demo.txt","r")
for i in range(5):
     print(fp.readline())

fp.close()
```

```
Hello

Hi

Bye

Excuse me

Avjo
```

04) WAP to find the longest word(s) in a file.

In [30]:
```
fp = open("demo.txt","r")
data = fp.readlines()

maximum = data[0]

for i in data:
    if len(maximum) < len(i):
        maximum = i

print(maximum)
```

```
ExcuseMe
```

05) WAP to count the no. of lines, words and characters in a given text file.

In [41]:
```
fp = open("demo.txt","r")
data = fp.readlines()
print(f"Lines: {len(data)}")

words = 0
char = 0
for i in data:
    words += len(i.split())
```

```
    char += len(i)


print("Words:",words)
print("Char:",char)
```

```
Lines: 5
Words: 9
Char: 46
```

06) WAP to copy the content of a file to the another file.

In [42]:
```
fp1 = open("demo.txt","r")
fp2 = open("new.txt","w")
fp2.write(fp1.read())
fp1.close()
fp2.close()
```

07) WAP to find the size of the text file.

In [44]:
```
fp = open("demo.txt","r")
data = fp.readlines()

char = 0
for i in data:
    char += len(i)

print("Size of File:",char)
```

```
Size of File: 46
```

08) WAP to create an UDF named frequency to count occurances of the specific word in a given text file.

In [46]:
```
def frequency(words):
    fp = open("demo.txt","r")
    c = 0
    for i in fp:
        c += i.split().count(words)
    print(c)

frequency("Hi")
```

```
2
```

09) WAP to get the score of five subjects from the user, store them in a file. Fetch those marks and find the highest score.

In [63]:
```
fp = open("new_1.txt","w")
li = []
for i in range(1,6):
    d = int(input(f"Enter marks:{i}"))
    li.append(d)
```

```
    fp.write(f"marks of sub{i} is {d}\n")

maximum  = max(li)
print("Maximum:",maximum)
```

```
Maximum: 100
```

10) WAP to write first 100 prime numbers to a file named primenumbers.txt

(Note: each number should be in new line)

In [ ]:
```
def is_prime(num):
    if num < 2:
        return False
    for i in range(2, int(num**0.5) + 1):
        if num % i == 0:
            return False
    return True


count = 0
num = 2

with open("primenumbers.txt", "w") as file:
    while count < 100:
        if is_prime(num):
            file.write(str(num) + "\n")
            count += 1
        num += 1

print("First 100 prime numbers written successfully.")
```

11) WAP to merge two files and write it in a new file.

In [7]:
```
file1 = open("demo.txt", "r")
file2 = open("new.txt", "r")
merged_file = open("merged.txt", "w")


merged_file.write(file1.read())
merged_file.write("\n")
merged_file.write(file2.read())

file1.close()
file2.close()
merged_file.close()

print("Files merged successfully into merged.txt")
```

```
Files merged successfully into merged.txt
```

12) WAP to replace word1 by word2 of a text file. Write the updated data to new file.

In [5]:
```
word1 = input("Enter word to be replaced: ")
word2 = input("Enter new word: ")

with open("input.txt", "r") as file:
    data = file.read()

updated_data = data.replace(word1, word2)

with open("output.txt", "w") as file:
    file.write(updated_data)

print("Word replaced successfully. Updated data written to output.txt")
```

```
Word replaced successfully. Updated data written to output.txt
```

13) Demonstrate tell() and seek() for all the cases(seek from beginning-end-current position) taking a suitable example of your choice.

In [4]:
```
with open("demo.txt", "w") as f:
    f.write("ABCDEFGHIJKLMNOPQRSTUVWXYZ")


with open("demo.txt", "r") as f:
    print("Initial position:", f.tell())

    data = f.read(5)
    print("Read:", data)
    print("Position after reading 5 chars:", f.tell())

    f.seek(10, 0)
    print("\nAfter seek(10,0) position:", f.tell())
    print("Reading next 5 chars:", f.read(5))
```

```
Initial position: 0
Read: ABCDE
Position after reading 5 chars: 5

After seek(10,0) position: 10
Reading next 5 chars: KLMNO
```

In [ ]: